

Liquidity, Activism Disclosure, and Takeover Premia

A Theory of Exit, Voice, and Corporate Control

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Activist Returns and the Takeover Channel

- **Brav et al., 2008:** Roughly 65% of activist hedge fund returns come from M&A events
- **Greenwood and Schor, 2009:** Activist targets are acquired at significant premia; activists profit primarily through the takeover channel
- **Collin-Dufresne and Fos, 2015:** Informed trading before Schedule 13D filings moves prices; activists trade strategically on private information

Central Question

How does market liquidity shape the takeover premia that minority shareholders receive when a blockholder can exit, hold, or engage?

Three Literatures, One Missing Piece

Exit / Voice

Edmans, 2009, Maug, 1998, Admati and Pfleiderer, 2009: liquidity enables exit-based governance and stake-building for intervention.

Microstructure & Feedback

Kyle, 1985, Back et al., 2018: informed trading, strategic liquidity, and price discovery.

Takeover & Real Effects

Edmans, Goldstein, et al., 2012: prices guide real decisions, including takeover entry.

The Gap

No model links all three: how does market liquidity, through its effect on information in prices, shape takeover premia for minority shareholders?

Contribution:

- (i) Non-monotone liquidity effect on premia
- (ii) Disclosure attenuation mechanism
- (iii) Premium wedge from tender mechanics
- (iv) Liquidity regulation $\equiv$ governance policy

Positioning: The Structural Activism Trio

- **Johnson and Swem, 2021** (JFE): dynamic *reputation* — the model's only state variable; accumulation collapsed to a lognormal cost draw; no trading, no disclosure margin.
- **Albuquerque et al., 2022** (JFE): structural 13D-vs-13G choice; engagement cost a *scalar*; 13D returns $\approx 3/4$ treatment — the market *capitalizes* activism at disclosure.
- **Celentano and Levine, 2025**: equilibrium activism + takeovers + discipline; estimate activism *lowers* bid premia 13.7% via board bargaining; no market microstructure.

This paper's lane

Liquidity, order-flow inference, and the disclosure rule are the trio's **missing state variables**. The premium they estimate as a constant is here an **equilibrium object of tender mechanics** ($\lambda = 1 - q(1 - \gamma)\psi$) — and the negative activism–premium estimate is the low- λ region of our model, with cross-sectional content $(q, \gamma, \psi, \kappa)$ their board-bargaining channel cannot index.

Disclosure Creates Two Information Regimes

Institutional facts:

- **US:** 5% threshold (Schedule 13D); window cut **10 calendar** → **5 business days** (eff. Feb 2024)
- **UK:** 3% threshold, 2-day window (FCA TR-1)
- Creates two observability regimes:

Quiet Voice ($D=0$)	Public Voice ($D=1$)
No disclosure triggered	Filing triggers $D=1$
Engagement <i>inferred</i> from X	Engagement <i>observed</i>
Inference depends on κ	Independent of κ

Model Translation

$$D = \mathbf{1}\{q = +1\}$$

Buy-and-engage $\Rightarrow$ public disclosure

Hold-and-engage $\Rightarrow$ no disclosure

Disclosure partitions equilibrium into **inference** ($D=0$, κ matters) vs. **observed** ($D=1$, κ irrelevant) regimes.

Alternative regimes ↗

Notation Guide

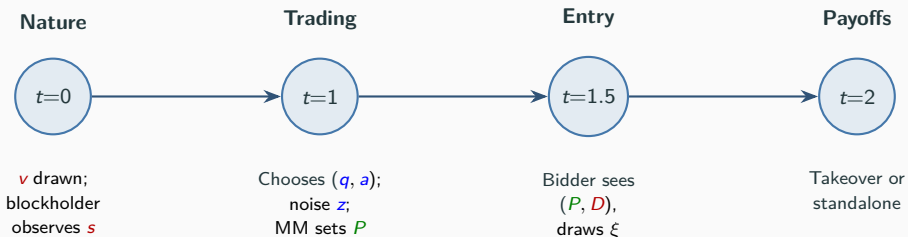
Choices / Actions	
$q \in \{-1, 0, +1\}$	Trade order
$a \in \{0, 1\}$	Engage?
$z \in \{-1, 0, +1\}$	Noise trade
Observables	
$X = q + z$	Order flow
$D = \mathbf{1}\{q = +1\}$	Disclosure
$s = v + \varepsilon$	Private signal
$v \sim N(\mu, \sigma_v^2)$	Fundamental

Equilibrium Objects	
$\pi(X, D)$	Posterior $\mathbb{P}(a=1 X, D)$
$P(X, D)$	Equilibrium price
$p(P, D)$	Bid probability
$m(X, D)$	Expected premium
Cutoffs (Prop. 1)	
k_1	Exit / Hold
k_0	Hold / Quiet Voice
k_D	Quiet / Public Voice

Parameters	
κ	Liquidity (noise)
ρ	Engage success prob.
$\Delta, \tilde{\Delta}$	Value-add ($\tilde{\Delta} = \rho\Delta$)
m_0, m_1	Premia (base, success)
$\tilde{m}$	$m_0 + \rho(m_1 - m_0)$
C_0, χ	Engage cost params
$\tilde{S}, K, \sigma_\xi$	Bidder entry
δ	Discount factor
Region probs	
$\omega_E, \omega_H, \omega_Q, \omega_P$	Action probs

Blue = choices Red = observables Green = outcomes

Model Timeline & Actions



	$a = 0$	$a = 1$
$q = -1$	Exit	–
$q = 0$	Hold	Quiet Voice
$q = +1$	–	Public Voice

Noise: $z \in \{-1, 0, +1\}$, $\mathbb{P}(z=0) = 1 - \kappa$, $\mathbb{P}(z = \pm 1) = \kappa/2$

Feedback: Bidder conditions entry on (P, D) , creating a price-to-entry loop.

Standing assumptions ↗

Engagement cost:

$$C(s) = C_0 \exp\left(-\chi \frac{s - \mu}{\sigma_s}\right)$$

Decreasing in signal $\Rightarrow$ strong-thesis activism.

Success: with probability ρ :

- Standalone value $+\Delta \Rightarrow \tilde{\Delta} = \rho\Delta$
- Premium rises: $m_0 \rightarrow m_1 \Rightarrow$
 $\tilde{m} = m_0 + \rho(m_1 - m_0)$

Fixed-point derivation ↗

Bidder entry:

$$p(P, D) = 1 - \Phi\left(\frac{\tilde{m}(X, D) + K - \bar{S} + P}{\sigma_\xi}\right)$$

Pricing fixed point:

$$P = \delta \left[(1 - \rho) \hat{V} + \rho(P + \tilde{m}) \right]$$

where $\hat{V} = \mathbb{E}[v|X, D] + \tilde{\Delta} \pi(X, D)$.

Regularity: $\delta \varphi(0)/\sigma_\xi < 1$ ensures unique P^* .

Bid deterrence ↗

Takeover comparative statics ↗

Equilibrium: Monotone Cutoff Strategy

Proposition 1 (Cutoff Structure)

In any PBE, cutoffs $k_1 \leq k_0 \leq k_D$:

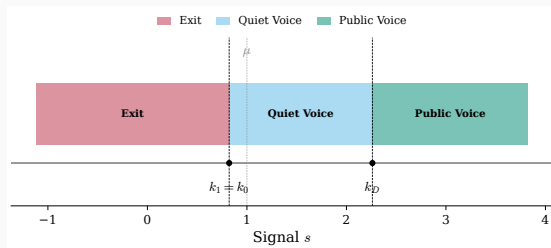
$s < k_1 \Rightarrow \text{Exit}$

$k_1 \leq s < k_0 \Rightarrow \text{Hold}$

$k_0 \leq s < k_D \Rightarrow \text{Quiet Voice}$

$s \geq k_D \Rightarrow \text{Public Voice}$

Key: Hold and Quiet Voice produce identical order-flow distributions $\Rightarrow$ market cannot distinguish them.



Equilibrium regions in signal space.

$k_0 = k_1$ at baseline (Hold collapsed).

[Proof sketch ↗](#)

Proposition 2 (Posteriors)

Disclosed: $\pi(X, 1) = 1 \ \forall X$. Non-disclosed ($D=0$, $p_0=1-\kappa$, $p_1=\kappa/2$):

$$\begin{aligned}\pi(1, 0) &= \frac{\omega_Q}{\omega_H + \omega_Q} \quad (\kappa\text{-free}) & \pi(0, 0) &= \frac{\omega_Q p_0}{(\omega_H + \omega_Q) p_0 + \omega_E p_1} \\ \pi(-1, 0) &= \frac{\omega_Q p_1}{(\omega_H + \omega_Q) p_1 + \omega_E p_0}\end{aligned}$$

Mechanism: $X \rightarrow \pi(X, D) \rightarrow \bar{m}(X, D) \rightarrow p(P, D) \rightarrow \Delta^{\min}$

κ enters at inference, **only on** $D=0$: rising κ compresses posteriors toward the prior.

Full posterior formulas ↗

Disclosed-branch invariance ↗

Proposition 3 (Price Decomposition)

$$P(X, D) = \delta \left(\underbrace{(1 - p^*) \mathbb{E}[v \mid X, D]}_{\text{fundamental}} + \underbrace{(1 - p^*) \tilde{\Delta} \pi}_{\text{standalone channel}} \right. \\ \left. + \underbrace{p^* (P + m_0)}_{\text{baseline takeover}} + \underbrace{p^* (\tilde{m} - m_0) \pi}_{\text{takeover channel}} \right)$$

Standalone channel:

Engagement raises expected standalone value by $\tilde{\Delta} \pi$. Weighted by $(1 - p^*)$: matters when no bid arrives.

Takeover channel:

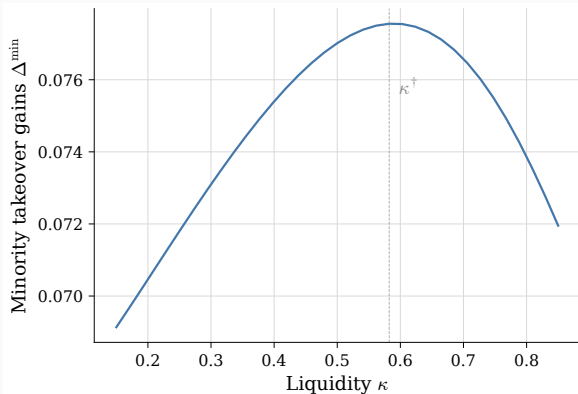
Engagement raises the expected premium wedge by $(\tilde{m} - m_0) \pi$. Weighted by p^* : matters when a bid arrives.

Both channels depend on $\pi(X, D)$: the inferred probability of engagement.

Pricing derivation ↗

Disclosed-branch invariance ↗

Main Result: Non-Monotone Liquidity Effect



Proposition 5

$\Delta^{\min}(\kappa)$ is **hump-shaped**: moderate κ uniquely maximizes minority takeover gains.

Scope: the GE clause holds as a *theorem* on a certified region, with a certified counterexample at its boundary (Theorem B).

Decomposition:

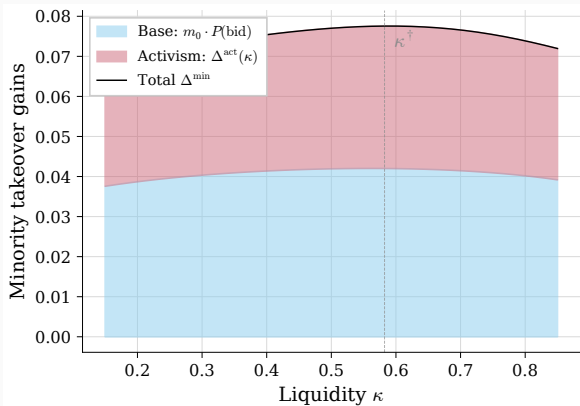
$$\Delta^{\min} = \underbrace{m_0 \cdot \mathbb{P}(\text{bid})}_{\text{base}} + \underbrace{(\tilde{m} - m_0) \mathbb{E}[\pi \cdot \mathbf{1}_{\text{bid}}]}_{\text{activism}}$$

Two forces:

- (i) $\kappa \uparrow \Rightarrow$ noise cover $\Rightarrow$ more bids
- (ii) $\kappa \uparrow \Rightarrow$ weaker inference $\Rightarrow$ lower π

At $\kappa^\dagger$: marginal benefit of noise cover = marginal cost of inference destruction.

Decomposition: Two Competing Forces



Sensitivity analyses ↗

Two Components

$$\Delta^{\text{min}} = \underbrace{m_0 \cdot \mathbb{P}(\text{bid})}_{\text{base}} + \underbrace{(\tilde{m} - m_0) \mathbb{E}[\pi \cdot \mathbf{1}_{\text{bid}}]}_{\text{activism}}$$

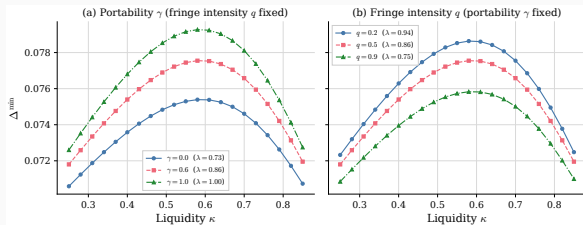
Base (dashed): monotone $\uparrow$ — noise $\Rightarrow$ lower prices $\Rightarrow$ more bids.

Activism (dotdash): hump-shaped — inference degrades as κ rises.

At $\kappa^\dagger$: activism peak drives the overall maximum.

Decomposition proof ↗

Theorem A: The Premium Wedge from Tender Mechanics



Appropriability is now derived

$$\lambda = 1 - q(1 - \gamma)\psi \in [0, 1]$$

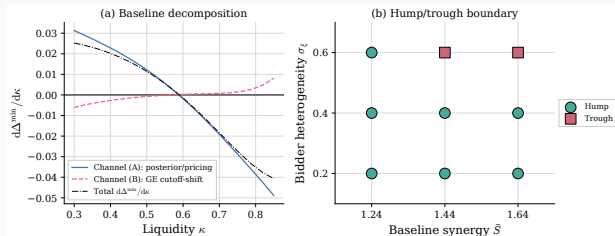
q : fringe-raid intensity · γ : improvement portability · ψ : pivotality

$m_1 - m_0 = (1 - \theta)\lambda\rho\Delta_{\text{eng}}$. A3 fails **only if** fringe raids are certain, fully superseding, and unblockable.

Negative premium estimates (Celentano and Levine, 2025) reconciled: $\bar{m}/P$ falls in activism evidence when λ is small.

The tender game ↗

Theorem B: When Is the Hump a Theorem?



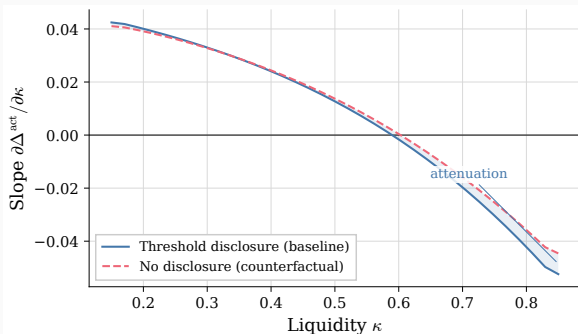
Certified region: GE cutoff-shift channel bounded *inversion-free* via the contraction modulus ($L \leq 0.84$): hump = **theorem** on $\kappa \in [0.35, 0.825]$.

Certified counterexample: at $\sigma_\xi=0.60$ channel (A) stays single-peaked, yet equilibrium $\Delta^{\min}$ is a **trough** — GE-dominance is false globally.

The boundary is economic: GE feedback wins exactly where bid entry is insensitive to premia.

How the region is certified ↗

Disclosure Attenuates the Liquidity Effect



Proposition 6 (Attenuation)

The slope $\partial \Delta^{\text{act}} / \partial \kappa$ is flatter under disclosure: the $D=1$ states contribute zero sensitivity.

The shaded region shows the attenuation gap: disclosure mechanically dampens κ -sensitivity.

Implication: Disclosure and liquidity are **substitutes** in shaping inference.

UK 3% threshold attenuates sensitivity relative to US 5%.

Alternative regimes ↗

Noisy rumors ↗

PE vs GE distinction ↗

Four testable predictions:

1. **Hump-shaped premia:** Quadratic κ term in premia regressions (SDC/CRSP)
2. **Disclosure attenuates:** Interact liquidity $\times$ UK/US; UK effect muted
3. **13D event study:** Abnormal return upon filing $\uparrow$ in illiquidity
4. **Bid deterrence:** Lower bid hazard for 13D filers, but higher conditional premia

GE disclosure analysis ↗

Welfare decomposition ↗

PE vs GE distinction ↗

GE Disclosure Trade-off (Prop. 7)

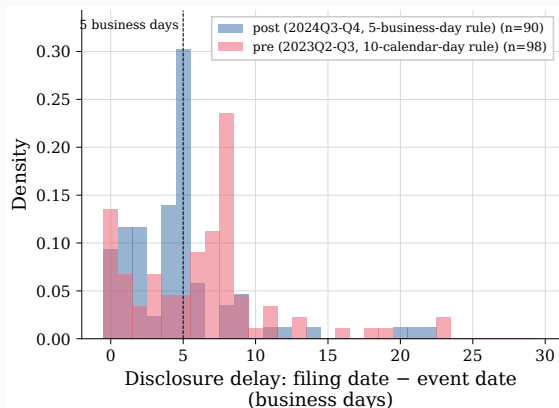
$$\frac{d\Delta^{\text{act}}}{d\tau} \approx \underbrace{\text{transparency}}_{+} + \underbrace{\text{deterrence}}_{-}$$

Strict disclosure can **paradoxically harm** minorities by destroying Quiet Voice.

C_0 low, κ moderate: transparency dominates.
 C_0 high: deterrence dominates.

Key: Liquidity regulation is implicitly governance policy.

Evidence I: The 2024 Acceleration Bit



Rule change: SEC release 33-11253 cut the Schedule 13D initial deadline from **10 calendar days** to **5 business days** (compliance 2024-02-05).

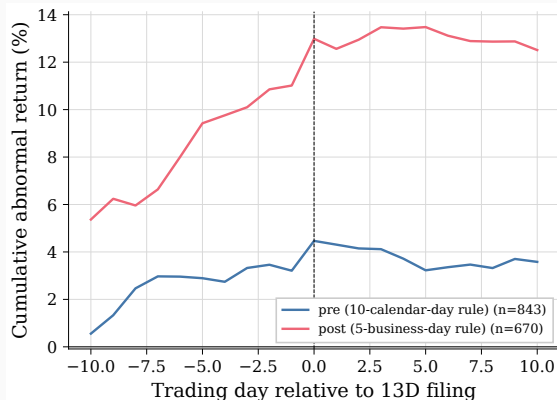
Fact 1: Filing delays compressed

Median event-to-filing delay falls **7 → 5** business days; the share filed within 5 business days more than doubles: **36% → 76%**.

EDGAR originals, 2023Q2–Q3 vs. 2024Q3–Q4 — the model's quiet-trading cover, shortened by law.

[Data & methods](#) ↗

Evidence II: Returns on the Modeled Margin



Design: market-model CARs around original 13D filings, 2022–2025; 1,513 events (843 pre, 670 post); *Post* = 5-business-day regime.

Fact 2: What the data say

Mean CAR[−1, +1] **doubles**:

0.85% → 1.71% ($\beta > 0$ as predicted; $t \approx 1.1$, n.s.).

Surprises concentrate in **liquid** names:

$\partial \text{CAR} / \partial \ln \text{Amihud} = -1.4\text{pp}$ ($t = -3.7$)
— the camouflage margin is first-order.

Post×illiquidity interaction: precise null.

Two-way clustered (firm, month); table in backup.

Summary

Mechanism

Liquidity changes what order flow reveals about hidden engagement. Disclosure creates observability vs. inference regimes.

Main Prediction

Minority takeover gains are **hump-shaped** in liquidity: bid incidence rises but the activism premium erodes.

Contribution

Unifies exit/voice, informed trading, and takeover literatures. Disclosure policy implicitly governs the liquidity–premium relationship. The premium wedge is derived from a free-rider tender game ($\lambda = 1 - q(1 - \gamma)\psi$; rationalizes the estimated negative activism–premium effect), and the GE clause of the hump is certified as a theorem on $[0.35, 0.825]$ with a genuine counterexample at $\sigma_\xi = 0.60$.

Extensions

Dynamic multi-round trading, multiple blockholders, endogenous information acquisition.

Thank you!

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Backup: Mathematical Foundations & Equilibrium Robustness

How do you establish that the cutoff ordering holds?

Proof sketch (Proposition 1): Single-crossing of payoffs in s .

Step 1: Exit vs. Hold. $B_E(s) - B_H(s)$ is decreasing in s : holding becomes relatively more attractive as v improves (higher terminal payoff, lower exit price relative to fundamental). $\Rightarrow \exists k_1$ with Exit optimal below.

Step 2: Hold vs. Quiet Voice. $B_H(s) - B_Q(s)$: engagement cost $C(s)$ is decreasing in s (strong-thesis activism), while the payoff uplift $\tilde{\Delta}$ and $\tilde{m} - m_0$ are constant. $\Rightarrow \exists k_0 \geq k_1$ with Quiet optimal above.

Step 3: Quiet vs. Public Voice. $B_Q(s) - B_P(s)$: Public Voice requires buying an additional share at price P , but gains double the terminal payoff. The cost of buying (P) is fixed in s while the benefit of a larger holding position increases with expected fundamental value. $\Rightarrow \exists k_D \geq k_0$ with Public optimal above.

The weak ordering $k_1 \leq k_0 \leq k_D$ permits region collapse (e.g., $k_0 = k_1$: no Hold region).

How do you know the equilibrium is unique?

Existence (Proposition 4): Brouwer's fixed-point theorem guarantees existence unconditionally. The cutoff mapping $T: (k_1, k_0, k_D) \mapsto (k'_1, k'_0, k'_D)$ maps a compact convex set to itself.

Uniqueness: Requires the spectral radius of the Jacobian DT to be < 1 (contraction condition). This is analytically intractable due to the nested fixed-point structure (prices depend on cutoffs, cutoffs depend on prices).

Numerical verification:

- Following Edmans, Goldstein, et al., 2015: verify numerically that the solver converges to the *same* cutoffs from arbitrary starting points
- Multi-start search with 4+ initial conditions yields identical solutions
- Convergence rate consistent with spectral radius $\ll 1$

Sufficient conditions: Bounded price-feedback (Assumption A5: $\delta\varphi(0)/\sigma_\xi < 1$) and non-pathological signal noise.

How does the market update its expectation of fundamentals?

Conditional means $\mathbb{E}[v \mid \text{action region}]$:

Let $\alpha_j = (k_j - \mu)/\sigma_s$. Using truncated normal expectations (Lemma 2):

$$\mu_E = \mathbb{E}[v \mid s < k_1] = \mu - \beta\sigma_s \cdot \frac{\phi(\alpha_1)}{\Phi(\alpha_1)} \quad (\text{inverse Mills ratio})$$

$$\mu_H = \mathbb{E}[v \mid k_1 \leq s < k_0] = \mu + \beta\sigma_s \cdot \frac{\phi(\alpha_1) - \phi(\alpha_0)}{\Phi(\alpha_0) - \Phi(\alpha_1)}$$

$$\mu_Q = \mathbb{E}[v \mid k_0 \leq s < k_D] = \mu + \beta\sigma_s \cdot \frac{\phi(\alpha_0) - \phi(\alpha_D)}{\Phi(\alpha_D) - \Phi(\alpha_0)}$$

$$\mu_P = \mathbb{E}[v \mid s \geq k_D] = \mu + \beta\sigma_s \cdot \frac{\phi(\alpha_D)}{1 - \Phi(\alpha_D)}$$

Bayesian mixing: $\mathbb{E}[v \mid X, D] = \sum_j \mathbb{P}(j \mid X, D) \mu_j$ over seven reachable (X, D) states.

Walk me through the pricing fixed-point solution.

For each information set (X, D) :

1. Compute $\pi(X, D)$ from Bayes' rule (Proposition 2)
2. Compute $\hat{V}(X, D) = \mathbb{E}[v \mid X, D] + \tilde{\Delta} \pi(X, D)$
3. Define $\bar{m}(X, D) = m_0 + (\tilde{m} - m_0) \pi(X, D)$
4. Solve the scalar fixed point:

$$P = \delta((1 - p(P, D)) \hat{V} + p(P, D)(P + \bar{m}))$$

$$\text{where } p(P, D) = 1 - \Phi\left(\frac{\bar{m} + K - \bar{S} + P}{\sigma_\xi}\right)$$

Closed form (rearranged): $P^* = \frac{\delta[(1 - p^*)\hat{V} + p^*\bar{m}]}{1 - \delta p^*}$

Regularity: $\delta \varphi(0)/\sigma_\xi < 1$ ensures the map $P \mapsto \delta[(1 - p(P))\hat{V} + p(P)(P + \bar{m})]$ is a contraction. At baseline: $0.95 \times 0.3989/0.40 = 0.947 < 1$. ✓

What are the full posterior formulas?

Disclosed branch ($D = 1$):

- $X \in \{0, 1, 2\}$ (since $q = +1$)
- $\pi(X, 1) = 1$ for all X
- $\mathbb{E}[v|X, 1] = \mu_P$ for all X
- Engagement fully observed; inference trivial

$p_0 = 1 - \kappa$, $p_1 = \kappa/2$:

$$\pi(-2, 0) = 0$$

$$\pi(1, 0) = \frac{\omega_Q}{\omega_H + \omega_Q}$$

$$\pi(0, 0) = \frac{\omega_Q p_0}{(\omega_H + \omega_Q) p_0 + \omega_E p_1}$$

$$\pi(-1, 0) = \frac{\omega_Q p_1}{(\omega_H + \omega_Q) p_1 + \omega_E p_0}$$

Non-disclosed ($D = 0$):

- $X \in \{-2, -1, 0, 1\}$ (since $q \in \{-1, 0\}$)
- $\pi(X, 0)$ via Bayes' rule
- Depends on κ through (p_0, p_1)

($X = -2$ requires $q = -1$: Exit only)

(independent of κ)

Partial derivatives (Section 5.4): $\frac{\partial \pi(1,0)}{\partial \kappa} = 0$, $\frac{\partial \pi(-1,0)}{\partial \kappa} \geq 0$, $\frac{\partial \pi(0,0)}{\partial \kappa} \leq 0$.

How do you derive the $\Delta^{\min}$ decomposition?

Goal: Prove $\Delta^{\min} = m_0 \cdot \mathbb{P}(\text{bid}) + (\tilde{m} - m_0) \mathbb{E}[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}]$.

Step 1: Definition. $\Delta^{\min}(\kappa) = \mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}]$ where $m^R(a) = m_0 + a(\tilde{m} - m_0)$.

Step 2: Conditional independence. Given (X, D) , the bid indicator $\mathbf{1}\{\text{bid}\}$ depends on $(P(X, D), D, \xi)$. Since $\xi \perp (v, a)$, the bid indicator is conditionally independent of (v, a) :

$$\mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\} \mid X, D] = \bar{m}(X, D) \cdot p(X, D)$$

Step 3: Iterated expectations. Taking $\mathbb{E}$ over (X, D) and substituting $\bar{m} = m_0 + (\tilde{m} - m_0)\pi$:

$$\Delta^{\min} = m_0 \cdot \mathbb{P}(\text{bid}) + (\tilde{m} - m_0) \cdot \mathbb{E}[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}] \quad \blacksquare$$

Key: The conditional independence step requires ξ (bidder shock) to be independent of (v, a) — the bid is determined by public information and idiosyncratic bidder heterogeneity alone.

What are the endpoint limits of minority gains?

Lemma 1 (Endpoint Behavior)

As $\kappa \uparrow 1$: $\Delta^{\text{act}}(\kappa) \rightarrow 0$, $\Delta^{\text{min}}(\kappa) \rightarrow m_0 \mathbb{P}(\text{bid})$.

As $\kappa \downarrow 0$: $\Delta^{\text{act}}(\kappa) > 0$ but suppressed below interior optimum.

Right endpoint ($\kappa \uparrow 1$):

- $\mathbb{P}(z=0) \rightarrow 0$: order flow uninformative
- $\pi(X, 0) \rightarrow$ unconditional prior
- Engagement return $< C(s)$ for all s
- Voice regions collapse: $\omega_Q + \omega_P \rightarrow 0$

Left endpoint ($\kappa \downarrow 0$):

- $\mathbb{P}(z=0) \rightarrow 1$: $X \rightarrow q$ a.s.
- Market perfectly deduces $q = 0$
- Engagement benefit bounded below $\Rightarrow \omega_Q > 0$ persists
- But full informativeness depresses $p(X, 0)$ via bid deterrence

$\Rightarrow$ The global maximum of $\Delta^{\text{min}}(\kappa)$ cannot be at either boundary.

How do you know the hump is strict?

Proof of Proposition 5 (Nonmonotonic Liquidity Effect):

Step 1: Continuity. $\Delta^{\min}(\kappa)$ is continuous on $[0, 1]$ (normal probabilities, Bayesian posteriors, and pricing are all continuous in κ).

Step 2: Weierstrass. Continuous on compact $\Rightarrow$ attains a global maximum

$$\kappa^{\dagger} = \arg \max_{\kappa \in [0,1]} \Delta^{\min}(\kappa).$$

Step 3: Interior maximizer. By Lemma 1, neither endpoint achieves the maximum:

$$\Delta^{\min}(\tilde{\kappa}) > \lim_{\kappa \downarrow 0} \Delta^{\min}(\kappa) \quad \text{and} \quad \lim_{\kappa \uparrow 1} \Delta^{\min}(\kappa) = m_0 \mathbb{P}(\text{bid})$$

$\Rightarrow \kappa^{\dagger} \in (0, 1)$: the hump is **strict** with an interior peak. ■

Note on shape: The Weierstrass argument guarantees at least one interior peak but does not rule out multiple local maxima (W-shape). The baseline calibration confirms a well-behaved single-peaked hump, demonstrating that the fundamental economic trade-off dominates.

What game pins the premium wedge?

Disagreement-node tender game. Players: fringe raider (entry cost $c_F \sim H$, synergy $S_F \sim G$), bloc (stake α , strategic), atomistic float (measure $1 - \alpha$); charter permits dilution φ of non-tendering shareholders (Grossman–Hart exclusion device).

Timing: raider enters and posts a uniform unrestricted offer $\hat{b}$ for control threshold $\tau \rightarrow$ float tenders atomistically (free-rides) $\rightarrow$ control transfers iff tenders $\geq \tau$; post-raid per-share value $w + \gamma a \Delta_{\text{eng}} + S_F$.

Equilibrium (with $\alpha < \tau$: the bloc alone cannot deliver control): free-rider floor $\hat{b}^* = w + \gamma a \Delta_{\text{eng}} + S_F - \varphi$; a *pivotal* bloc ($1 - \alpha < \tau$) blocks floor raids unless $S_F - \varphi \geq (1 - \gamma) a \Delta_{\text{eng}}$. On the blocking set the raider can still top up to the bloc's reservation — but such raids pay the bloc *exactly* its outside option, so the threat point (and λ) is unchanged.

Appropriability as an equilibrium output

$$\lambda = 1 - q(1 - \gamma)\psi, \quad \psi = \frac{1}{(1 - \gamma)\Delta_{\text{eng}}} \int_0^{(1 - \gamma)\Delta_{\text{eng}}} \mathbb{P}(S_F - \varphi \geq t) dt \in [0, 1]$$

($\psi = 1$ when non-pivotal.) $\lambda > 0$ fails **only if** $q = 1$, $\gamma = 0$, and blocking is impossible or worthless — so A3 is a theorem, not an assumption. Wedge: $m_1 - m_0 = (1 - \theta)\lambda\rho\Delta_{\text{eng}}$. Residue: one-shot fringe, exogenous (φ, τ) .

How is the certified region actually certified?

Problem: the hump needs the GE cutoff-shift channel $B(\kappa)$ not to overturn the fixed-cutoff channel (A). Bounding B requires $\|dk/d\kappa\|$ — normally the *unknown* equilibrium path derivative.

Inversion-free bound: the cutoff map $F = I - T$ inherits the A6 contraction modulus L of the best-response map T , so by the Neumann series

$$\left\| (I - D_k T)^{-1} \right\|_{\infty} \leq \frac{1}{1 - L} \implies \left\| \frac{dk}{d\kappa} \right\|_{\infty} \leq \frac{\|\partial_{\kappa} T\|_{\infty}}{1 - L},$$

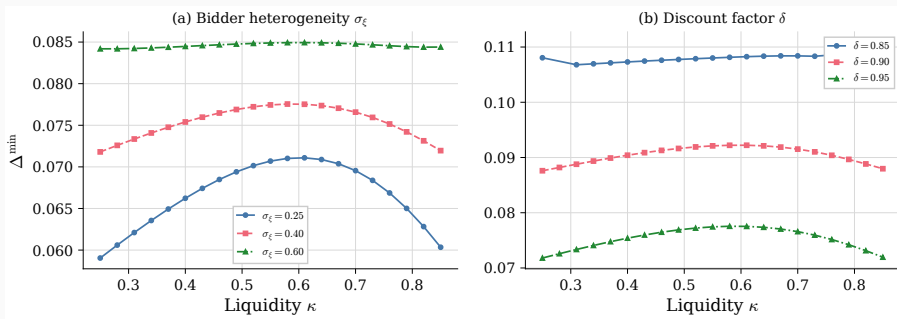
computable from *one* equilibrium solve plus one finite-difference column — no Jacobian inversion, no path knowledge.

At the baseline: $L \leq 0.836 < 1$ along the whole path; channel-(A) peak at $\kappa \approx 0.60$; certification margins hold at every off-ball grid point (ball integral 2.5×10^{-5} vs. margin 4.7×10^{-3}) $\Rightarrow$ exact-IFT region $[0.30, 0.85]$, inversion-free region $[0.35, 0.825]$.

And the counterexample is certified too: at $\sigma_{\xi} = 0.60$, troughs at interior $\kappa \in \{0.275, 0.35, 0.475\}$ with channel (A) single-peaked in every cell and $\int |B|$ exceeding the (A)-amplitude (e.g. 0.021 vs. 0.0069) — GE dominance is genuinely non-removable.

Backup: Sensitivity & Comparative Statics

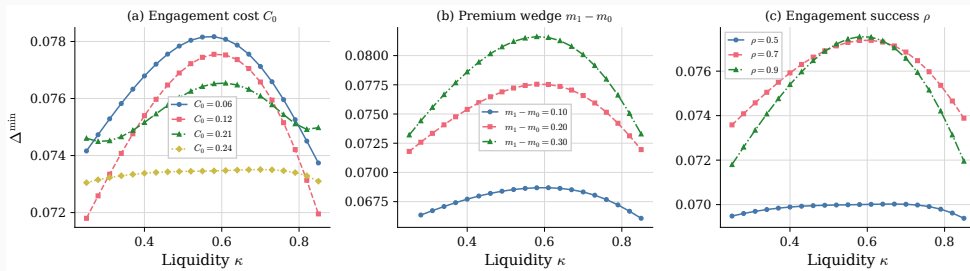
Sensitivity: Discount Factor & Bidder Shock



Left: Lower δ shifts curve up; hump flattens. **Right:** Hump robust to bidder variance σ_ξ . Green dotted: $\kappa^\dagger$.

← Slide 10: Main Result

Sensitivity: Cost, Premium Wedge & Success Probability



Left: Higher C_0 flattens but preserves nonmonotonicity. **Center:** Larger $m_1 - m_0$ raises conditional premia but deters entry. **Right:** Lower ρ compresses vertically; $\rho = 0.9$ reflects selective activists.

← Slide 10: Main Result

Does engagement deter bids?

Bid probability: $p(X, D) = 1 - \Phi\left(\frac{\tilde{m}(X, D) + K - \bar{S} + P}{\sigma_\xi}\right)$ where $T \equiv \frac{\tilde{m} + K - \bar{S} + P}{\sigma_\xi}$.

Result 1: Higher prices deter bids.

$$\frac{\partial p}{\partial P} = -\frac{\phi(T)}{\sigma_\xi} < 0$$

The supremum $\sup_P |\partial p / \partial P| = \phi(0) / \sigma_\xi$ is the binding term in Assumption A5.

Result 2: Higher inferred engagement deters bids (when $\tilde{m} > m_0$).

$$\frac{\partial p}{\partial \pi} = -\frac{(\tilde{m} - m_0) \phi(T)}{\sigma_\xi} < 0$$

Policy implication: Disclosed activism ($\pi = 1$) deters bids more strongly than inferred activism ($\pi < 1$), since the full premium $\tilde{m}$ enters the bidder's cost. This is why $p(X, 1) < p(X, 0)$ in the baseline table.

How do synergies and entry costs affect bids?

Fix any (X, D) . Define $T \equiv (\bar{m} + K - \bar{S} + P)/\sigma_\xi$, so $p = 1 - \Phi(T)$.

Result (a): Higher synergies increase bids.

$$\frac{\partial p}{\partial \bar{S}} = \frac{\phi(T)}{\sigma_\xi} > 0 \quad (\text{since } \partial T / \partial \bar{S} = -1/\sigma_\xi)$$

Result (b): Higher entry costs reduce bids.

$$\frac{\partial p}{\partial K} = -\frac{\phi(T)}{\sigma_\xi} < 0 \quad (\text{since } \partial T / \partial K = +1/\sigma_\xi)$$

Unified structure: All three derivatives $(\bar{S}, K, P)$ share the same $\phi(T)/\sigma_\xi$ factor. Differences are purely in sign, driven by how each parameter enters the bidder's standardized threshold T .

Note: These are partial equilibrium results (equilibrium prices held fixed). In GE, higher $\bar{S}$ also raises equilibrium prices, partially offsetting the direct effect through the price-entry feedback loop.

Why are prices flat on the disclosed branch?

Claim: $P^*(x, 1) = P^*(x', 1)$ and $p(x, 1) = p(x', 1)$ for all $x, x' \in \{0, 1, 2\}$.

Step 1: Identification. $D = 1 \Leftrightarrow$ Public Voice: $a = 1$ a.s. and $X = 1 + z$ with $z \in \{-1, 0, 1\}$.

Step 2: Independence. Since $z \perp (v, s, \xi)$, conditioning on $X = x$ given $D = 1$ only reveals $z = x - 1$ — no information about v beyond what $D = 1$ already implies:

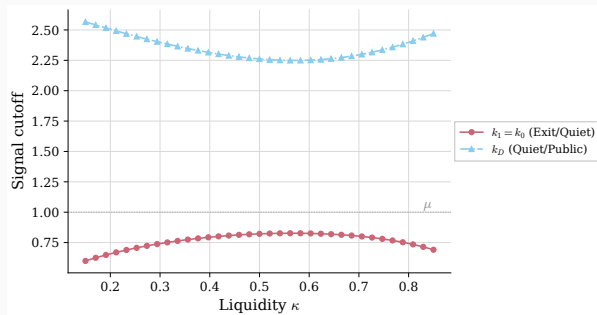
$$\mathbb{P}(v \in B \mid X=x, D=1) = \mathbb{P}(v \in B \mid D=1) \quad \forall x \in \{0, 1, 2\}$$

$\Rightarrow \pi(x, 1) = 1, \mathbb{E}[v \mid x, 1] = \mu_P$ for all x .

Step 3: Fixed-point uniqueness. Since $\hat{V}(x, 1)$ and $\bar{m}(x, 1) = \tilde{m}$ are constant, the pricing fixed point has the same RHS for all x . By A5, $P^*(x, 1) = P^*(x', 1)$. ■

Consequence: The disclosed component of Δ^{act} is κ -invariant — this is the foundation of Proposition 6 (disclosure attenuation).

How do equilibrium cutoffs move with liquidity?



- $k_1 = k_0$ throughout (Hold collapsed at baseline)
- k_D tracks the Quiet/Public Voice boundary
- Weak ordering $k_1 \leq k_0 \leq k_D$ preserved across all κ

Backup: Disclosure Extensions & Information Regimes

Full disclosure vs. no disclosure benchmarks

Full Information (FI): Market observes (q, a) directly.

- $\pi_{FI} = a$ (deterministic); inference channel shuts down
- Activism premium deterministic: $\bar{m} = m_0$ or $\tilde{m}$
- Order flow uninformative about engagement
- *Upper bound* on disclosure informativeness

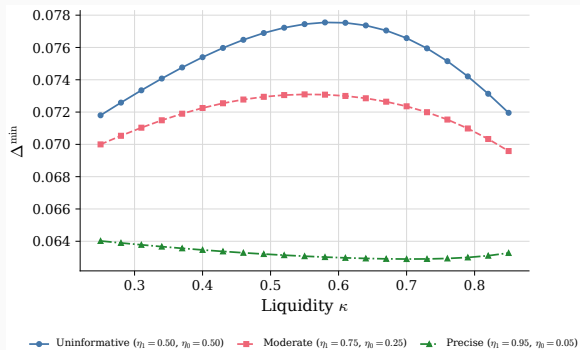
No Disclosure (ND): Market conditions only on X (no D signal).

- $\pi_{ND}(X)$ pools over all (q, a) states producing X
- Activism premium more κ -sensitive (no $D = 1$ anchor)
- *Lower bound* on disclosure informativeness

Ordering: $\text{Var}_{\kappa}[\Delta_{FI}^{\text{act}}] = 0 < \text{Var}_{\kappa}[\Delta_{\text{baseline}}^{\text{act}}] < \text{Var}_{\kappa}[\Delta_{ND}^{\text{act}}]$

These are the upper and lower bounds on the disclosure attenuation effect.

How do noisy rumors affect the mechanism?



Extension (Section 8.3): Augment market's information with binary rumor $R \in \{0, 1\}$:

- $\mathbb{P}(R=1 \mid a=1) = \eta_1$ (true positive)
- $\mathbb{P}(R=1 \mid a=0) = \eta_0$ (false positive)

Updated posterior:

$$\pi(X, 0, R) = \frac{\omega_Q \cdot p_Z \cdot \eta_R^Q}{\sum_j \omega_j \cdot p_Z \cdot \eta_R^j}$$

As $\eta_1 \rightarrow 1, \eta_0 \rightarrow 0$: converges to FI within $D=0$ states.

Systematic **flattening** as rumor precision increases: additional information about quiet engagement substitutes for disclosure, further attenuating the liquidity effect.

GE effect of stricter disclosure?

Proposition 7 (GE Disclosure Trade-off)

$$\frac{d\Delta^{\text{act}}}{d\tau} \approx \underbrace{\left. \frac{\partial \Delta^{\text{act}}}{\partial \tau} \right|_{\text{fixed}}}_{\text{Transparency (+)}} + \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial k_D} \cdot \frac{dk_D}{d\tau}}_{\text{Deterrence (—)}}$$

Transparency dominates:

- Low C_0 , moderate κ
- Strict disclosure improves welfare

Deterrence dominates:

- High C_0
- Destroys Quiet Voice, *harms* minorities

Policy: SEC 13D filing window reforms must account for deterrence, not just transparency.

← Slide 12: Policy Implications

PE vs. GE: Two disclosure effects

Proposition 6 (Partial Equilibrium):

- **Fix** cutoffs (k_1, k_0, k_D)
- Vary κ through inference only
- Disclosed states ($D=1$): κ -invariant
- Inferred states ($D=0$): weight $1 - \omega_P$

$\Rightarrow |\partial \Delta^{\text{act}} / \partial \kappa|$ **decreases** as ω_P rises.

Key distinction: PE shows *why* disclosure attenuates (mechanical: $D=1$ states are κ -free). GE shows *whether* stricter disclosure helps. Low C_0 + moderate κ : transparency dominates. High C_0 : deterrence dominates.

Proposition 7 (General Equilibrium):

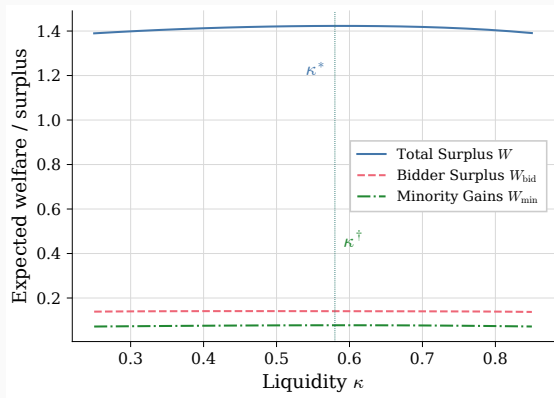
- **Allow** $k_D = k_D(\tau)$ to adjust
- Total derivative decomposes:

$$\frac{d\Delta^{\text{act}}}{d\tau} \approx \underbrace{(+)}_{\text{transp.}} + \underbrace{(-)}_{\text{deter.}}$$

- Stricter τ destroys Quiet Voice
- Blockholder may exit rather than disclose

Backup: Welfare

Distributional or total surplus?



Welfare: $W(\kappa) = W_{min} + W_B + W_{bid}$

Prices/premia are transfers. Surplus = synergies + improvements – costs.

Tension: $\kappa^\dagger$ maximizes extraction, deterring marginal efficient takeovers. $\Rightarrow \kappa^* \neq \kappa^\dagger$.

Why use expected minority takeover gains as the welfare metric?

Grossman & Hart (1980): Dispersed minorities benefit only through the takeover *premium*.

$\Delta^{\min}$ captures:

1. *Probability* a bid arrives (prices embed inference)
2. *Premium* conditional on bid (inferred vs. observed)
3. Interaction of these channels through κ

Empirical: activist returns $\approx$ takeover premia (Brav et al., 2008). $\Delta^{\min} = ex\ ante$ expected CAR.

← Slide 10: Main Result

Backup: Calibration & Numerical Details

What are your baseline parameter values and why?

Parameter	Value	Rationale
μ (prior mean)	1.00	Normalization
σ_v (value std)	0.50	Moderate uncertainty
σ_ε (noise std)	0.50	$\beta = 0.5$ (equal weight)
δ (discount)	0.95	Near-term horizon
κ (liquidity)	0.50	Interior baseline
C_0 (base cost)	0.12	Moderate engagement
χ (cost sensitivity)	0.50	Gradual decay

Derived: $\sigma_s = 0.707$, $\beta = 0.5$, $\tilde{m} = 0.28$, $\tilde{\Delta} = 0.225$.

Regularity check: $\delta/\sigma_\xi = 0.95/0.40 = 2.375 < 1/\varphi(0) \approx 2.507$. ✓ (Assumption A5 satisfied.)

Parameter	Value	Rationale
ρ (success prob)	0.90	Selective activists
Δ (value-add)	0.25	25% of μ
m_0 (base premium)	0.10	10% of μ
m_1 (success premium)	0.30	30% of μ
$\bar{S}$ (synergy)	1.44	Interior bid prob
K (entry cost)	0.15	Moderate friction
σ_ξ (shock vol)	0.40	Bidder heterogeneity

← Slide 6: Engagement & Bidder Entry

Show me the full equilibrium outcomes at baseline.

Cutoffs: $k_1 = k_0 \approx 0.82$, $k_D \approx 2.26$.

D	X	$P(X, D)$	$\pi(X, D)$	$p(X, D)$	$\tilde{m}(X, D)$
0	-2	0.84	0.00	0.81	0.10
0	-1	1.06	0.41	0.55	0.17
0	0	1.23	0.74	0.33	0.23
0	1	1.39	1.00	0.17	0.28
1	0	1.90	1.00	0.01	0.28
1	1	1.90	1.00	0.01	0.28
1	2	1.90	1.00	0.01	0.28

Key: (i) $D=1$: $\pi = 1$, flat prices; (ii) $D=0$: inference gradient; (iii) $D=1$ has *much lower* bid probability — disclosure deters bids through the price run-up and the full premium $\tilde{m}$.

What are the within-regime liquidity effects on posteriors?

Partial derivatives (Appendix B.7):

1. $\frac{\partial \pi(1,0)}{\partial \kappa} = 0$ — $X=1, D=0$ pins $q=0$; mix of Hold/Quiet depends on cutoffs, not κ .
2. $\frac{\partial \pi(0,0)}{\partial \kappa} \leq 0$ — Rising κ dilutes inference of Quiet Voice; compresses $\pi(0,0)$ toward prior.
3. $\frac{\partial \pi(-1,0)}{\partial \kappa} \geq 0$ — Rising κ makes $X=-1$ more likely from Hold/Quiet with bad noise, raising π .

Net effect on Δ^{act} : probability-weighted sum across states yields the hump.

← Slide 8: Proposition 2

What are the standing assumptions?

	Assumption	Where Used
A1	Normal fundamentals: $v \sim N(\mu, \sigma_v^2)$	Posterior mean, conditional expectations
A2	Normal signal noise: $s = v + \varepsilon$, $\varepsilon \sim N(0, \sigma_\varepsilon^2)$	Bayesian updating, cutoff regions
A3	Discrete noise trading: $z \in \{-1, 0, +1\}$	Order flow inference, posterior formulas
A4	Exponential engagement cost: $C(s) = C_0 e^{-\chi s}$	Single-crossing (Prop. 1)
A5	Bounded feedback: $\delta\varphi(0)/\sigma_\xi < 1$	Price fixed-point uniqueness
A6	Cutoff mapping is a contraction	Equilibrium uniqueness (Prop. 4)
A7	Bidder observes (P, D) but not (X, s, v, a)	Information structure, entry rule

Note: A5 is verifiable ex ante from primitives. A6 is verified numerically (following Edmans, Goldstein, et al., 2015). All other assumptions are standard.

Backup: Data & Evidence

How are the two facts constructed?

Sample. All *original* Schedule 13Ds on EDGAR (amendments excluded), full-text indexes 2022Q1–2025Q4. Form-type matching is boundary-aware: EDGAR renamed the form SC 13D → SCHEDULE 13D in Dec. 2024, and naive fixed-width reads conflate originals with amendments (and would drop 2025 entirely).

Fact 1 (delays). Event date from the cover-page label (or the structured XML tag in the post-2024 regime); delay = business days from event to filing; windows 2023Q2–Q3 vs. 2024Q3–Q4 straddle the 2024-02-05 compliance date.

Fact 2 (CARs). Subject CUSIP parsed from cover pages with check-digit validation and label-proximity voting (≈ 80 – 90% coverage); filings accepted after 16:00 ET shift to the next trading day. Market-model abnormal returns (CRSP daily; estimation 220 trading days ending 30 before the event, min. 100 obs); $CAR[-1, +1]$ and $CAR[-10, +1]$.

Covariates & inference. Amihud illiquidity over $[-250, -30]$, run-up $[-60, -11]$, $\ln$ market cap at -30 ; US common stock only (CIZ share-class filter); OLS with CGM two-way clustered SEs (firm $\times$ calendar month); year-quarter FE absorb *Post* in the saturated spec.